1. a) The generated auxiliary problem:

$$\underset{x,y}{\text{minimize}} \ \mathbf{1}^T y$$

subject to $Ax + y = b$

$x, y \geq 0$

∴ For original variable x_j:

Because the constraint matrix A is $m \times n$,

∴ The number of basic variables in the solution is no more than m.

But the non-basic variables could also be non-zero

∴ x^* could have more than m non-zero entries.

∴ (x^*, y^*) could have more than m non-zero entries.

b) False.

Phase I could only determine whether the original problem is feasible or not.

The unboundness is determined in Phase II

(c) False.

A counterexample:

The original problem: minimize $x_1 + 2x_2$

subject to $x_1 + x_2 + s_1 = 3$

$x_1 + s_2 = 3$ ∴ $m=1, n=2$, $A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

$x_2 + s_3 = 3$.

$x_1, x_2, s_1, s_2, s_3 \geq 0$

∴ $x^* = (0, 3, 0, 3, 0)$, which is unique with 2 positive components

(d) True.

If the basic solution is feasible,

it is a basic feasible solution (BFS)

∴ It is an extreme points of the set $\{x : Ax = b, x \geq 0\}$.

∴ If a basic solution is not an extreme point of the set $\{x : Ax = b, x \geq 0\}$, it is not feasible.

2.

$$\begin{array}{lrl} \text{maximize} & x_1 + 2x_2 + 3x_3 + 4x_4 & \\ \text{subject to} & x_1 + x_2 + x_3 & \leq 1 \\ & x_3 - x_4 & \leq 2 \\ & x_2 - 2x_3 + 4x_4 & \leq 3 \\ & x_1, x_2, x_3, x_4 & \geq 0. \end{array}$$

$\Rightarrow x_1 + x_2 + x_3 + s_1 = 1$

$x_3 - x_4 + s_2 = 2$

$x_2 - 2x_3 + 4x_4 + s_3 = 3$

$x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0$

⇒ minimize $c^T x$, $c^T = [-1 \ -2 \ -3 \ -4 \ 0 \ 0 \ 0]$,

subject to $Ax = b$.

$x \geq 0$ $A = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 4 & 0 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

∴ For basis $\{5, 6, 7\}$, with BFS $x^T = [0 \ 0 \ 0 \ 0 \ 1 \ 2 \ 3]$

B	-1	-2	-3	-4	0	0	0	0
5	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	0	1	-2	4	0	0	1	3

New basis $\{1, 6, 7\}$, with $x^T = [1 \ 0 \ 0 \ 0 \ 0 \ 2 \ 3]$

B	0	-1	-2	-4	0	0	0	1
1	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	0	1	-2	4	0	0	1	3

The new basis: $\{1, 6, 7\}$, with BFS $x^T = [1 \ 0 \ 0 \ 0 \ 0 \ 2 \ 3]$

B	1	0	-1	-4	2	0	0	2
2	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	-1	0	-3	4	-1	0	1	3

The new basis: $\{2, 6, 7\}$, with BFS $x^T = [0 \ 1 \ 0 \ 0 \ 1 \ 5]$

B	2	1	0	-4	3	0	0	3
3	1	1	1	0	1	0	0	1
6	-1	-1	0	-1	-1	1	0	1
7	2	3	0	4	2	0	1	5

The new basis: $\{3, 4, 6\}$, with BFS: $[0 \ 0 \ 1 \ \frac{5}{4} \ 0 \ \frac{9}{4} \ 0]$

B	4	4	0	0	5	0	0	8
3	1	1	1	0	1	0	0	1
6	$-\frac{1}{2}$	$-\frac{1}{4}$	0	0	$-\frac{1}{2}$	1	$\frac{1}{4}$	$\frac{3}{4}$
4	2	3	0	4	2	0	1	5

∴ $x = [0 \ 0 \ 1 \ \frac{5}{4} \ 0 \ \frac{9}{4} \ 0]^T$, the optimal value is $3 + 5 = 8$

3.

$$\begin{array}{lrl} \text{minimize} & x_1 + x_2 + 2x_4 & \\ \text{subject to} & x_1 - x_2 & \geq 1 \\ & x_1 + x_2 - x_3 - x_4 & \leq 2 \\ & x_2, x_3 & \geq 0. \end{array}$$

Transfer into standard form:

∵ $x_1 - x_2 \geq 1$ ∴ $x_1 \geq 1 + x_2$

∵ $x_2 \geq 0$ ∴ $x_1 \geq 0$

∵ $x_1 + x_2 - x_3 - x_4 \leq 2$

∴ $x_4 \geq x_1 + x_2 - x_3 - 2$

∴ ⇒ minimize $x_1 + x_2 + 2x_4$

subject to $x_1 - x_2 - s_1 = 1$

$x_1 + x_2 - x_3 - x_4 + s_2 = 2$

$x_1, x_2, x_3, s_1, s_2 \geq 0$

∴ phase I: minimize $s_3 + s_4$

subject to $x_1 - x_2 - s_1 + s_3 = 1$

$x_1 + x_2 - x_3 - x_4 + s_2 + s_4 = 2$

$x_1, x_2, x_3, s_1, s_2, s_3, s_4 \geq 0$

∴ $c^T = [0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1]$, $A = \begin{bmatrix} 1 & -1 & 0 & 0 & -1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

∴ If we choose basis $\{7, 8\}$, the BFS $x = [0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 2]$

B	2	0	1	1	-1	0	0	3
7	1	-1	0	0	-1	0	1	1
8	1	1	-1	-1	0	1	0	2

∴ The new basis: $\{1, 8\}$, with new BFS $x = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]$

B	0	2	1	1	-1	-1	2	0	-1
1	1	-1	0	0	-1	0	1	0	1
8	0	2	-1	-1	1	1	-1	1	1

∴ The new basis: $\{1, 2\}$, with BFS $x = [\frac{3}{2} \ \frac{1}{2} \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$

∴

B	0	0	0	0	0	0	1	1	0
1	1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

∴ The original problem is feasible.

$x = [\frac{3}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0]$

Phase II:

$c^T = [1 \ 1 \ 0 \ 2 \ 0 \ 0 \ 0]$, $A = \begin{bmatrix} 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 1 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

∴ $\tilde{c}^T = c^T - c_B^T A_B^{-1} A = [1 \ 1 \ 0 \ 2 \ 0 \ 0 \ 0] - [1 \ 0 \ -\frac{1}{2} \ -\frac{1}{2} \ \frac{1}{2} \ \frac{1}{2} \ \frac{1}{2}] = [0 \ 1 \ \frac{1}{2} \ \frac{5}{2} \ -\frac{1}{2} \ -\frac{1}{2} \ -\frac{1}{2}]$

∴ The new basis: $\{1, 6\}$, with BFS $x = [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0]$

B	0	2	0	2	1	0	-1
1	1	-1	0	0	-1	0	1
2	0	2	-1	-1	1	1	1

∴ The optimal value is 1, $x = [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0]$

4.

$$\begin{aligned} \text{minimize} \quad & x_2 - 4x_3 + x_5 \\ \text{subject to} \quad & x_1 + x_2 + x_3 + x_4 = 1 \\ & x_2 + x_3 + x_4 - x_5 = -1 \\ & x_1 + 2x_3 - x_5 \geq 0 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

$\Rightarrow$ minimize $x_2 - 4x_3 + x_5$

subject to $x_1 + x_2 + x_3 + x_4 = 1$

$x_2 + x_3 + x_4 - x_5 = -1$

$x_1 + 2x_3 - x_5 - s_1 = 0$

$x_1, x_2, x_3, x_4, x_5, s_1 \geq 0$

$\therefore c^T = [0\ 1\ -4\ 0\ 1]$, $A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 \\ 1 & 0 & 2 & 0 & -1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

phase I: minimize $s_2 + s_3 + s_4$

subject to $x_1 + x_2 + x_3 + x_4 + s_2 = 1$

$-x_2 - x_3 - x_4 + x_5 + s_3 = 1$

$x_1 + 2x_3 - x_5 - s_1 + s_4 = 0$

$x_1, x_2, x_3, x_4, x_5, s_1, s_2, s_3, s_4 \geq 0$

$\therefore A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & -1 & -1 & 0 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $c^T = [0\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 1]$

If we choose basis $\{7, 8, 9\}$, BFS $x^T = [0\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 0]$

B	-z	x_1	x_2	x_3	x_4	x_5	s_1	s_2	s_3	rhs
7	1	1	1	1	1	0	0	1	0	1
8	0	-1	-1	-1	1	0	0	1	0	1
9	1	2	0	0	-1	-1	0	0	1	0

$\rightarrow$ New basis: $\{1, 8, 9\}$. BFS: $x^T = [1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ -1]$ $\Rightarrow$ infeasible

For $\{1, 5, 6\}$, $x^T = [1\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0]$, $s_2 + s_3 + s_4 = 0$

$\therefore s_2 + s_3 + s_4 = 0$

$\therefore$ The original problem is feasible

Phase II:

Start with $\{1, 5, 6\}$,

$c^T = [0\ 1\ -4\ 0\ 1\ 0]$

$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 1 & 0 \\ 1 & 0 & 2 & 0 & -1 & -1 \end{bmatrix}$, $A_B^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\therefore \bar{c}^T = [0\ 1\ -4\ 0\ 1\ 0] - [0\ 1\ 0] \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix} \cdot A$

$= [0\ 1\ -4\ 0\ 1\ 0] - [0\ 1\ 0] \cdot A$

$= [0\ 2\ -3\ 1\ 0\ 0]$

$\therefore$ New basis: $\{3, 5, 6\}$, $x^T = [0\ 0\ 1\ 0\ 2\ 0]$

$\therefore$

B	-z	x_1	x_2	x_3	x_4	x_5	rhs
3	1	1	1	1	0	0	1
5	-1	-2	-2	-2	1	0	2
6	1	3	1	3	0	1	0

$\therefore$ The optimal value is 2, $x = [0\ 0\ 1\ 0\ 2\ 0]$

5. a) $\because$ There is a "-1" on the first row

$\therefore$ The current solution is not optimal

To make the solution feasible, $x_B \geq 0$

$\therefore \eta \geq 0$

(b) $\because$ The LP is unbounded

$\therefore$ There exists one column with all non-positive entries and negative reduced costs

$\therefore$ ① For column 1: $\alpha < 0$, $\gamma < 0$

② For column 6: $\beta < 0$, but $1 \geq 0$, impossible

③ For column 7: $\delta < 0$

$\therefore \eta \geq 0$, $\alpha < 0$, $\gamma < 0$

or $\eta \geq 0$, $\delta < 0$

(c) For column 3: Impossible.

For column 6: If $\frac{\eta}{1} \cdot \beta = \eta\beta = -1$

For column 7: $\frac{1}{\delta} = 1 \Rightarrow \delta = 1$

$\therefore \eta \geq 0$, $\eta\beta = -1$ or

$\eta \geq 0$ $\delta = 1$.

(d) ① If $\alpha \geq 0$ and $\beta \geq 0$ and $\delta \geq 0$

$\Rightarrow \alpha - \frac{1}{\delta} \geq 0$, $1 - \frac{1}{\delta} \geq 0$

$\therefore \delta \geq 1$ $\alpha \geq 1$

$\therefore \alpha \geq 1$, $\delta \geq 1$, $\beta \geq 0$, $\eta \geq 0$ ✓

② $\alpha < 0$, $\eta \geq 0$, $\gamma \leq 0$ $\Rightarrow$ unboundness, ✓

③ $\alpha \geq 0$, $\beta < 0$ $\rightarrow$ Opposite on Column 6 $\Rightarrow$ Impossible.

④ $\alpha \geq 0$, $\beta \geq 0$, $\delta < 0$ $\Rightarrow$ Unboundness, ✓

$\therefore \alpha \geq 1$, $\delta \geq 1$, $\beta \geq 0$, $\eta \geq 0$ or

$\alpha < 0$, $\eta \geq 0$, $\gamma \leq 0$ or

$\alpha \geq 0$, $\beta \geq 0$, $\delta < 0$